VoiceOps 2026 Capstone Report

ZentroMart customer support agent | Audit snapshot: 25 October 2026 | Dataset reference date: 20 October 2026

What was built

A four-step support agent: Step 1 prompting with Gemini; Step 2 document-grounded retrieval with source IDs; Step 3 conversation memory, verified order lookup and ticket creation; Step 4 Gnani-only speech-to-text and text-to-speech. The local and hosted web interfaces reuse the same Python agent as the PowerShell workflow. Optional OpenAI-compatible embeddings and the in-memory vector index remain unchanged. Synthetic capstone data only.

Evaluation snapshot

Saved automated string/source screening results. Steps 1 and 2 used the same historical question input; the current evaluator defaults to `asr_text`, so these are not scores from a fresh `asr_text` run. No new Gemini evaluation was run in this audit. Human review is needed for paraphrases, dates/numbers, and Hindi/Hinglish quality.

Stage	Cases	Required facts	Evidence / tool-call checks
Step 1 - prompt only	52	8/49 (16.3%)	Evidence 0/47 (no RAG); calls 52/52
Step 2 - grounded RAG	52	42/49 (85.7%)	Evidence 47/47; calls 52/52
Step 3 - conversations, post-fix saved run	96 turns	59/77 (76.6%)	Evidence 96/96; calls 96/96
Step 4 - synthetic voice smoke, earlier run	3 turns + silence	Manual listening required	Gnani path and unclear-audio retry exercised

Within the saved question run, Step 2 improved required-fact screening over Step 1 and matched all 47 scored evidence checks. The post-fix conversation snapshot improved expected tool-call checks from 76/96 to 96/96 and evidence checks from 45/96 to 96/96. These literal checks are screening signals, not semantic grading.

Findings, fixes, and limits

Earlier misses involved guessed or lost order IDs, missing verification/policy evidence, and a missed human handover. The agent was updated to bind lookups to a caller-supplied or pending order ID, retrieve failed-verification guidance, escalate overdue verified refunds, and report ticket creation only after the tool confirms success. The remaining evaluation task is a quota-permitting Step 1/2 run on `asr_text`, followed by manual review of borderline answers.

Gnani remains the only STT/TTS provider. A prior synthetic batch smoke and silent-audio recovery case passed; a later browser microphone check returned “No speech detected,” so intelligible physical-microphone capture and voice quality remain unverified in that check. The 29.71-second demo shows a grounded clothing-return answer with source IDs, a follow-up, and a tool-confirmed ticket handover. Offline tests: 36 run, 35 passed, 1 skipped because local socket connections are blocked. Dataset validation passed all nine CSVs, 21 knowledge files, relations, order math, practice evidence, and verification gates.

Submission and reproducibility

Public repository: https://github.com/vanshdixit1/zentromart-voiceops
Hosted web app: https://zentromart-voiceops.vercel.app
PowerShell batch command: `python batch_run.py <audio-folder> --output sample_outputs`
Local web app: `python web_app.py`, then open `http://127.0.0.1:8000`. Bonus work includes a shared-agent web interface, encrypted hosted sessions, source/chunk outputs, and portfolio architecture visuals. Never include `.env`, API keys, virtual environments, or a dataset copy in the submission ZIP.